Complete schematic - sheet 1

Every component terminal is shown. Equal net labels are connected; use connection IDs in the CSV for physical routing.

E1 DESIGN HOLD | Physical connections unapproved | Generic ports require actual-part mapping | 02 Nov 2026

BT1 | 3S LiPo / connector

INVENTORY / VERIFY

11.1 V nominal / 3300 mAh

PACK+	PACK_RAW
PACK-	GND
BAL_CELL0	CELL0
BAL_CELL1	CELL1
BAL_CELL2	CELL2
BAL_CELL3	CELL3

TB_GND | Common ground distribution

ADDITIONAL / MISSING OR SELECT

Conductor / terminal rating TBD

BUS	GND

S2 | Latching E-stop / 3 isolated NC contacts

ADDITIONAL / MISSING OR SELECT

SELECT PART

COIL_COM	COIL_PG2
COIL_NC	COIL_STOP_CHAIN
SENSE_COM	V3V3
SENSE_NC	SENSE_STOP_OK
EN_COM	ENABLE_5V
EN_NC	EN_DRIVER_5V

Z_COIL | Coil suppression network

ADDITIONAL / MISSING OR SELECT

SELECT FOR K1

A1	COIL_HOLD
A2	GND

Z_L | Local driver-bus transient clamp

ADDITIONAL / MISSING OR SELECT

Standoff / clamp / energy TBD

+	LEFT_SUPPLY
-	GND

F0 | main pack fuse / holder

ADDITIONAL / MISSING OR SELECT

DC A / interrupt rating TBD

IN	PACK_RAW
OUT	PACK_FUSED

P1 | 3S high-side protection assembly

ADDITIONAL / MISSING OR SELECT

SELECT PART

IN+	PACK_ISOLATED
OUT+	PACK_PROTECTED
GND	GND
CELL0	CELL0
CELL1	CELL1
CELL2	CELL2
CELL3	CELL3

S3 | Momentary motor-power ARM

ADDITIONAL / MISSING OR SELECT

NO; rating TBD

COM	COIL_STOP_CHAIN
NO	COIL_HOLD

OV1 | 5V rail 1 OVP/OCP and supervision assembly

ADDITIONAL / MISSING OR SELECT

SELECT / VERIFY

IN+	V5_MCU_RAW
OUT+	V5_MCU
GND	GND
PG_COM	COIL_FUSED
PG_NO	COIL_PG1

Z_R | Local driver-bus transient clamp

ADDITIONAL / MISSING OR SELECT

Standoff / clamp / energy TBD

+	RIGHT_SUPPLY
-	GND

S0 | Master load isolator

ADDITIONAL / MISSING OR SELECT

DC rating TBD

COM	PACK_FUSED
OUT	PACK_ISOLATED

TB_POS | Protected positive distribution

ADDITIONAL / MISSING OR SELECT

Conductor / terminal rating TBD

BUS	PACK_PROTECTED

K1 | DC motor contactor

ADDITIONAL / MISSING OR SELECT

CONTACT / COIL TBD

MAIN_COM	PACK_PROTECTED
MAIN_NO	MOTOR_SWITCHED
A1	COIL_HOLD
A2	GND
AUX_HOLD_COM	COIL_STOP_CHAIN
AUX_HOLD_NO	COIL_HOLD
AUX_SENSE_COM	SENSE_STOP_OK
AUX_SENSE_NO	ESTOP_OK

OV2 | 5V rail 2 OVP/OCP and supervision assembly

ADDITIONAL / MISSING OR SELECT

SELECT / VERIFY

IN+	V5_LOGIC_RAW
OUT+	V5_LOGIC
GND	GND
PG_COM	COIL_PG1
PG_NO	COIL_PG2

Z_A | Local driver-bus transient clamp

ADDITIONAL / MISSING OR SELECT

Standoff / clamp / energy TBD

+	AUX_SUPPLY
-	GND

Red +supply | Black GND | Blue control | Green sensor | Orange motor | Thick = high-current | Dashed = missing/blocked

Net labels join across sheets. Crossings alone do not join. No component footprint / connector orientation is implied.